

When connecting any two devices, whether by cable or wirelessly, care should be taken such that both sender and receiver encode, transmit, and decode data in a similar fashion.

Wireless networking has several standards. The 802.11a standard transmits at 5.0 GHz and can move up to 54 megabits of data per second (around 6.75 MBps transfer speeds). The original Wi-Fi protocol 802.11 used a DSSS (direct sequence spread spectrum) modulation technique. In DSSS, a noise signal is multiplied along with the signal and scattered across a wide band of frequencies. The noise signal is a random sequence of + (plus) and - (minus) ones, synchronized between both sender and receiver. The frequency of the original signal is multiplied, spreading the energy to a wider band. The 802.11a improves on this and uses OFDM (Orthogonal Frequency Division Multiplexing), a more efficient coding technique that splits that radio signal into several sub-signals before they reach a receiver. This greatly reduces interference.

The 802.11b standard is the slowest, but is also the least expensive standard. 802.11b transmits in the 2.4 GHz band of the spectrum. It can handle only up to 11 megabits per second (as little over one MBps) and it uses complementary code keying (CCK) modulation to improve speeds. Complementary codes are sets of finite sequences of equal length. They are generated such that the number of pairs of identical elements with any given separation in one sequence is equal to the number of pairs of unlike elements having the same separation in the other sequences.

The 802.11g standard transmits at 2.4 GHz, like 802.11b, but it's a lot faster as it can handle up to 54 megabits of data per second. 802.11g is also faster because it uses the same OFDM coding as 802.11a.

802.11n is the newest standard that is due to become widely available. The standard significantly improves the speed and range of Wi-Fi devices. For instance, although 802.11g theoretically moves 54 megabits of data per second, it only achieves real world speeds of about 24 megabits of data per

While the 802.11b is the slowest, the newest version 802.11n is the fastest and will be available soon